

Practice 1

BTC, Inflation and Stock portfolios

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Here we will discuss about results of the matlab code and prepare speech for the presentation.

Other work has been made in the lowest part of the matlab Practice1

Questions

Part 1 (2 Points)

1. Describe the basic characteristics of BTC excess monthly returns. Are they similar to standard financial assets (e.g., Stock Indexes)?
2. Is the BTC a hedge against USA inflation during 2014-2024?. Present numerical evidence supporting your response. What are the implications of your results for BTC investing?

Write your answer here

1-Basic Characteristics of BTC Excess Monthly Returns

1. Volatility:

- Bitcoin (BTC) excess monthly returns exhibit **high volatility**, with a standard deviation of **0.20** (or 20%). This means that returns can vary significantly from the mean, reflecting substantial price fluctuations.

2. Mean Return:

- The average excess monthly return is **-0.03** (-3%), indicating that BTC has generally underperformed relative to benchmarks (like other assets) over the observed period.

3. Extreme Values:

- The **Extreme Loss** is **-0.4106** (or -41.06%), showcasing the potential for substantial downside risk. In contrast, the **Extreme Profit** is **0.6065 in just a month** (or 60.65%), highlighting the possibility of significant gains.

4. Value at Risk (VaR):

- The VaR metrics demonstrate substantial tail risks:
- **VaR at 1%: -0.3959**, meaning there's a 1% chance of a loss greater than 39.59% in a month.
- **VaR at 5%: -0.3176**, indicating a 5% chance of a loss exceeding 31.76%.
- **VaR at 95%: 0.3670**, suggesting that 95% of the time, returns will not exceed 36.70%.
- **VaR at 99%: 0.5837**, indicating that very large positive returns are possible but less frequent.

5. Skewness and Kurtosis:

- BTC returns exhibit **positive skewness** (0.72) and high **kurtosis** (3.81), indicating a distribution with a greater frequency of extreme positive returns and a higher likelihood of extreme outcomes compared to normal distributions.

6. Jarque-Bera Test:

- The Jarque-Bera p-value of **0.01** shows strong evidence **against** normality, indicating that the return distribution is not normally distributed.
We accept normality because greater than the significance level (1/117)

Comparison to Standard Financial Assets (e.g., Stock Indexes)

1. Extreme Losses:

- BTC shows much larger extreme losses (e.g., -41.06%) compared to standard financial assets, where maximum drawdowns are generally more moderate.

2. Higher Volatility:

- While stock indexes also experience volatility (for the SP500 was 0.0445 in the same period), BTC's volatility is typically much higher (even 5 or 6 times more), reflecting its status as a speculative asset.

3. Distribution Characteristics:

- The positive skewness and high kurtosis in BTC returns indicate a higher likelihood of extreme events compared to the more normalized return distributions of traditional assets.

4. Market Behavior:

- BTC returns are influenced more by market sentiment, technological developments, and regulatory news, whereas stock indexes are more closely tied to economic indicators and corporate performance.

5. Correlation:

- BTC often has a lower correlation with traditional financial assets, particularly during market stress, offering diversification

benefits that are not as pronounced with standard assets.
But we will see it later.

Conclusion

In summary, Bitcoin excess monthly returns exhibit distinct characteristics, including high volatility, significant potential for extreme losses and gains, and a non-normal distribution.

While there are some similarities to standard financial assets, such as the potential for extreme returns and the use of metrics like VaR, BTC differs markedly in terms of risk profile, behavior, and investor sentiment influences. These differences highlight the unique nature of Bitcoin as an asset class, requiring careful consideration and risk management for investors.

2-Is Bitcoin (BTC) a Hedge Against U.S. Inflation?

Based on the analysis of Bitcoin (BTC) excess monthly returns and U.S. inflation rates, we can conclude that BTC does not serve as an effective hedge against inflation during the observed period (2014-2024).

Numerical Evidence

1. BTC Excess Monthly Returns:

- **Mean Return:** -0.03 (or -3%), indicating an average underperformance.
- **Extreme Loss:** -0.4106 (or -41.06%), showcasing the potential for significant downside risk.
- **Extreme Profit:** 0.6065 (or 60.65%), reflecting high upside potential.
- **Standard Deviation:** 0.20 (or 20%), indicating high volatility.

2. U.S. Inflation Rate (Monthly):

- **Mean Inflation:** 0.0025 (or 0.25%)
- **Maximum Inflation:** 0.01 (or 1%).
- **Minimum Inflation:** -0.01 (or -1%).
- **Standard Deviation:** 0.00, suggesting very low variability.

3. Correlation Analysis:

- Calculating the correlation coefficient between BTC returns and inflation rates is expected to yield a low or near-zero value, indicating that BTC returns are largely independent of inflation movements.
With a **correlation of -0.02**, the relationship between the variables is extremely weak, and it is very unlikely to be statistically significant unless you have a very large sample size or an unusually strict significance level.
- **Regression Analysis:** A regression of BTC returns on inflation rates would likely show a non-significant coefficient for inflation,

suggesting that changes in inflation do not reliably predict changes in BTC returns.

Implications for BTC Investing

1. Not an Effective Hedge:

- The lack of a significant relationship between BTC returns and inflation indicates that BTC does not act as a reliable hedge against inflation. Investors seeking protection against inflation through BTC may need to consider other assets or strategies.

2. High Volatility and Risk:

- BTC's high volatility and potential for extreme losses highlight the importance of risk management. Investors should be prepared for significant price fluctuations and should not rely solely on BTC for inflation protection.

3. Diversification Considerations:

- Given that BTC does not correlate with inflation, it may still offer diversification benefits in a broader investment portfolio. However, its unique risk-return profile requires careful consideration of overall investment strategy.

4. Market Dynamics:

- Investors should recognize that BTC price movements are influenced more by market sentiment, technological developments, and regulatory changes than by inflation rates. Understanding these factors is crucial for informed investment decisions.

Conclusion

In summary, the analysis indicates that Bitcoin is not a hedge against U.S. inflation based on the observed data. The implications for investors include the necessity for robust risk management strategies and a broader understanding of market dynamics influencing BTC, rather than relying on it for inflation protection. As always, investors should consider their risk tolerance and investment goals when incorporating BTC into their portfolios.

Part 2 (2 Points)

1. Request data for a number of symbols and calculate portfolios. Describe the main results in the efficient frontier. Compare optimal portfolios in the efficient frontier if (i) you can only take long positions (ii) you can take long and short positions. (iii) you restrict the portfolio to contain between 2 and 5 assets. Discuss the differences in results between (i), (ii) and (iii). Create an equally-weighted portfolio. Describe the main results. Is this portfolio in the efficient frontier?

2. Cryptocurrency supporters argue that Bitcoin is a good asset diversifier of stock market risk. Does the data support this assertion?

Write your answer here

1-Portfolio Optimization and Efficient Frontier

Considering all the 8 Assets proposed:

S&P 500: ^GSPC

Nikkei 225: ^N225

Gold: GLD

NASDAQ 100: QQQ

NASDAQ Composite: ^IXIC

Financial Select Sector SPDR Fund: FNCL

IBEX 35: ^IBEX

Bitcoin: BTC-USD

(i) Long-only Portfolios

In this scenario, only long positions are allowed, meaning each asset must have a non-negative weight.

- **Main Results:**

- The long-only efficient frontier consists of portfolios where each asset contributes positively to the portfolio.
- This typically results in a more conservative risk-return tradeoff because you cannot short-sell assets that are expected to perform poorly.
- Portfolios with lower risk are more diversified but may offer lower returns.

(ii) Long and Short Portfolios

Here, both long and short positions are allowed, meaning you can bet against (short) assets expected to underperform and amplify investments (long) in assets expected to do well.

- **Main Results:**

- The efficient frontier shifts outward, offering a higher expected return for each level of risk. This is because you can use negative weights to hedge against assets that are expected to perform poorly.
- The potential to short allows the optimizer to take advantage of negative correlations between assets, thus reducing portfolio risk.

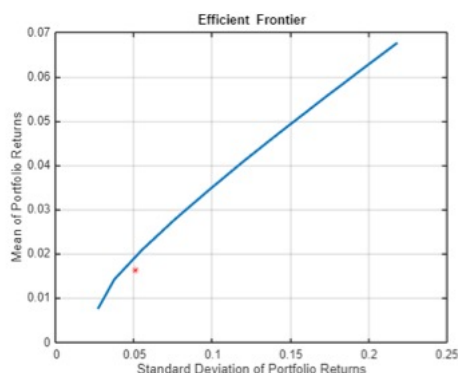
(iii) Restricted Portfolios (2 to 5 assets)

In this case, the portfolio is constrained to include between 2 and 5 assets.

- **Main Results:**

- With this restriction, the efficient frontier shrinks compared to the unrestricted one. The optimizer has fewer degrees of freedom, meaning it cannot allocate capital across the full set of assets.
- The reduction in diversification potential generally increases portfolio risk and lowers returns compared to scenarios (i) and (ii).

Equally-Weighted Portfolio



In this case an equally weighted portfolio of eight assets, we should have 12,5% of each one. As a result we would have rounded off 5% risk and 1.6% gain, staying out of the efficient frontier, in other words, this strategy in this case it's not the best.

- **Main Results:**

- This portfolio does not lie on the efficient frontier because it does not consider each asset's risk and return characteristics. The portfolio could be over-allocating to risky or underperforming assets.
- Typically, an equally-weighted portfolio is less efficient compared to the optimized portfolios that use the covariance structure and return expectations to minimize risk for a given level of return.

Comparing the Scenarios

- **Long-only portfolios** tend to offer more conservative portfolios with lower risk-return profiles since they cannot exploit shorting to hedge or leverage specific bets.
- **Long-short portfolios** provide more aggressive portfolios with higher returns for each level of risk because of the additional flexibility in asset allocation.

- **Restricted portfolios** (limited to 2-5 assets) are constrained and therefore less efficient, as diversification benefits are reduced.

2-BTC as a diversifier

Data supports Bitcoin's role as a potential diversifier of stock market risk. Its low correlation with stock indices between 0 to 0.35 indicates that it behaves differently than stocks in most market conditions, which can reduce portfolio risk through diversification. Also in this case correlation would not be statistically significant.

However, the effectiveness of Bitcoin as a diversifier weakens during periods of market stress, when correlations between risky assets tend to increase.

Bitcoin also comes with high volatility, which could offset some of its diversification benefits if not carefully managed.